UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 001 · Introduction to partial differentiation · compiled 15 October 2026

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1 Short notes

1 · Functions of two or more variables

Quantities depending on several variables need functions of several variables (area $A(l,b)$, volume $v(r,h)$, potential $V(x,y,z,t)$, weather, pandemics). $z=f(x,y)$ assigns to each $(x,y)\in D$ a unique real number; D is the domain, the set of values the range. The graph is a surface.

2 · Limit of a function of two variables

$\lim_{(x,y)\to(a,b)} f = L$ if every path to (a,b) gives L. Polynomial: substitute. $\frac{0}{0}$: try paths ($y=0$, $y=x$, $y=mx$, $y=x^2$); two different values ⇒ no limit. $\frac{2xy}{3x^2+y^2}$: 0 on $y=0$, $\frac{1}{2}$ on $y=x$.

3 · Continuity of a function of two variables

Continuous at (a,b): $f(a,b)$ defined, limit exists, $\lim f = f(a,b)$; no hole or break in the surface. Not continuous ≠ no limit: $\frac{3x^2y}{x^2+y^2}$ has limit 0 but $f(0,0)$ undefined.

4 · Partial derivatives: history, definition and notation

Partial derivative: derivative in one variable, others constant. $z_x = \lim_{h\to 0}\frac{f(x+h,y)-f(x,y)}{h}$; notation $\frac{\partial z}{\partial x} = z_x = f_x$. Leibniz (product rule), Legendre 1786, Jacobi 1841. All differentiation rules hold.

5 · Second-order and mixed partial derivatives

z_{xx}, z_{yy}, $z_{xy} = \frac{\partial}{\partial y}(z_x) = \frac{\partial^2 z}{\partial y \partial x}$, z_{yx}. Clairaut: $z_{xy} = z_{yx}$ when the second partials are continuous. Subscripts left to right, ∂ right to left.

2 Keyword index

19 terms; the number in brackets is the topic.

Bivariate function (1). $z = f(x, y)$, two independent variables.

Clairaut's theorem (5). $z_{xy} = z_{yx}$ if the second partials are continuous.

Continuity (3). $\lim f = f(a, b)$; no hole or break.

Domain (1). The set D of pairs (x, y).

First principles (4). $z_x = \lim_{h \to 0} \frac{f(x+h,y)-f(x,y)}{h}$.

Indeterminate form (2). $\frac{0}{0}$ on substitution; try paths.

Legendre (1786), Jacobi (1841) (4). Created and reintroduced the symbol ∂.

Leibniz (4). First used partial differentiation (product rule).

Limit (2). Same value L along every path to (a, b).

Mixed partial (5). z_{xy}, z_{yx}.

Multivariable function (1). A quantity depending on two or more variables.

Order of subscripts (5). Left to right = order of differentiation.

Partial derivative (4). Derivative in one variable, others constant.

Path (2). A curve (line, parabola, ellipse) along which $(x, y) \to (a, b)$.

Range (1). The set of values f takes on.

Removable discontinuity (3). Limit exists, $f(a, b)$ undefined or different.

Second-order partial (5). z_{xx}, z_{yy}.

Surface (1). Graph of $z = f(x, y)$.

Two-path test (2). Two different path values $\Rightarrow$ no limit.

3 Rapid recall

1	Define a function of two variables.	A rule assigning to each $(x, y) \in D$ a unique real number $f(x, y)$.
2	Graph of $z = f(x, y)$?	A surface in three dimensions.
3	When does $\lim_{(x,y)\to(a,b)} f$ exist?	When every path to (a, b) gives the same value.
4	How do you show a limit does not exist?	Find two paths giving different values.
5	$\lim_{(x,y)\to(2,1)}(x^2 - 2xy + 3y^2 - 4x + 3y - 6)$?	-8 (and -6 at $(2, -1)$).
6	$\frac{2xy}{3x^2+y^2}$ along $y = mx$?	$\frac{2m}{3+m^2}$: no limit at the origin.
7	Three conditions for continuity?	$f(a, b)$ defined, limit exists, limit $= f(a, b)$.
8	Is $\frac{3x^2y}{x^2+y^2}$ continuous at $(0, 0)$?	No ($f(0, 0)$ undefined), though the limit is 0.
9	Not continuous $\Rightarrow$ no limit?	No.
10	Definition of $\frac{\partial z}{\partial x}$?	$\lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$, y constant.
11	Who created and who reintroduced ∂?	Legendre (1786); Jacobi (1841).
12	Earliest use of partial differentiation?	Leibniz's product rule.
13	$\frac{\partial^2 z}{\partial y \partial x}$ in subscripts?	z_{xy} (x first).
14	Clairaut's theorem?	$f_{xy} = f_{yx}$ if the second partials are continuous.
15	$f = e^{x^2+y^2+z^2}$: f_{xx} at $(-1, 1, -1)$?	$6e^3$.
16	n for $v = r^n(3\cos^2\theta - 1)$?	$n = 2$ or -3.
17	$u = x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$: u_{xy}?	$\frac{x^2-y^2}{x^2+y^2}$.

Mnemonic (study aid). *"Subscripts walk forward, partials walk back."* z_{xy}: x then y; $\frac{\partial^2 z}{\partial y\,\partial x}$: read from the right, x then y.

4 Last 10 minutes before the exam

Top facts

1. $z = f(x, y)$: each $(x, y) \in D$ gives one real number; the graph is a surface.
2. Limit: the same value along every path; two paths with different values ⇒ no limit.
3. $\frac{2xy}{3x^2+y^2}$: 0 along $y = 0$, $\frac{1}{2}$ along $y = x$ ⇒ no limit at the origin.
4. Continuity: $f(a, b)$ defined, limit exists, $\lim f = f(a, b)$. Not continuous does not mean no limit.
5. $z_x = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$: differentiate in x, everything else constant.
6. Notation $\frac{\partial z}{\partial x} = z_x = f_x$; ∂ by Legendre (1786), reintroduced by Jacobi (1841).
7. Clairaut: $f_{xy} = f_{yx}$ when the second partials are continuous.
8. $\frac{\partial^2 z}{\partial y \partial x} = z_{xy}$: subscripts left to right, ∂ right to left.

Top formulas

E4.1 · Partial derivatives from the definition

ESA ★★

$$\frac{\partial z}{\partial x} = \lim_{h\to 0} \frac{f(x+h, y) - f(x, y)}{h}$$
$$\frac{\partial z}{\partial y} = \lim_{h\to 0} \frac{f(x, y+h) - f(x, y)}{h}$$

GIVES	z_x (y held constant) and z_y (x held constant).
APPLIES	When the limits exist.
NOTE	Only one variable is moved by h; the other stays fixed.

S001·15

E4.2 · Notation for first-order partials

ESA ★★

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x, \qquad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y$$

GIVES	The equivalent symbols.
APPLIES	$z = f(x, y)$.
NOTE	The symbol is ∂ (“partial”, often read “dou”), not δ or d.

S001·15 · T001b 00:11:41 · T001b 00:13:05

E5.2 · Mixed partials ESA ★★

$$z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y\,\partial x} = f_{xy}, \qquad z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x\,\partial y} = f_{yx}$$

GIVES The two crossed derivatives.

APPLIES Differentiate in one variable, then the other.

NOTE Subscripts: left to right. ∂ notation: right to left.

S001·16 · S001·17 · T001b 00:18:41

E5.3 · Clairaut's theorem ESA ★★★

$$f_{xy} = f_{yx} \qquad \left(\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}\right)$$

GIVES The order of differentiation does not matter.

APPLIES f_{xy} and f_{yx} continuous near the point.

NOTE Called “a very important theorem” in the lecture; it is why Taylor's series has $2hkf_{xy}$.

S001·17 · T001b 00:19:57 · T001b 00:20:10

E2.2 · Two-path test (limit does not exist) ESA ★

$$\lim_{\text{path 1}} f \neq \lim_{\text{path 2}} f \implies \lim_{(x,y)\to(a,b)} f(x,y) \text{ does not exist}$$

GIVES A proof that the limit does not exist.

APPLIES Substitution gives $\frac{0}{0}$.

NOTE Lines $y = mx$ through the origin are the usual first try: if the answer depends on m, the limit does not exist.

S001·10 · T001a 00:17:42

E3.1 · Continuity at (a, b) ESA ★

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

GIVES The test for continuity.

APPLIES Three conditions: $f(a,b)$ defined, limit exists, both equal.

NOTE One variable: $\lim_{x\to c} f(x) = f(c)$.

S001·11 · T001b 00:00:38